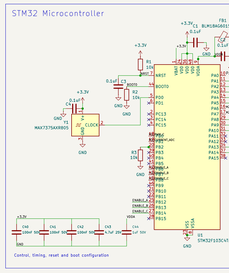
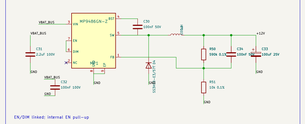
STM32 Microcontroller



Control, timing, reset and boot configuration

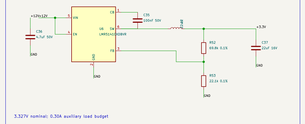
Auxiliary Power Supply



EN/SYNC linked; internal EN pull-up

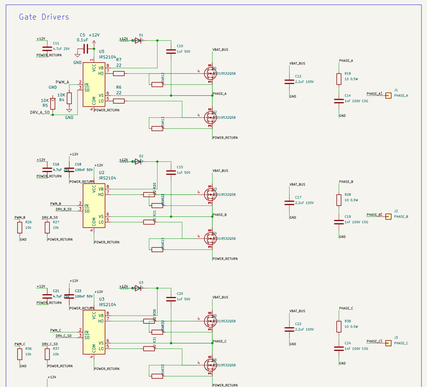
12V hysteretic buck; output ESR and ripple need bench verification

Nominal 12V × 0.2A = 2.1 × 90%/10A)



3.327V nominal: 0.36A auxiliary load budget

Gate Drivers

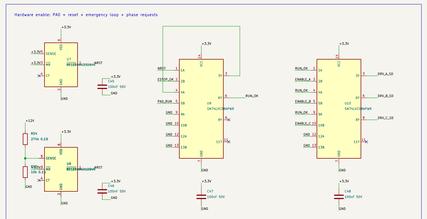


Gate-drive interfaces for the three motor phases

POWER_RETURN

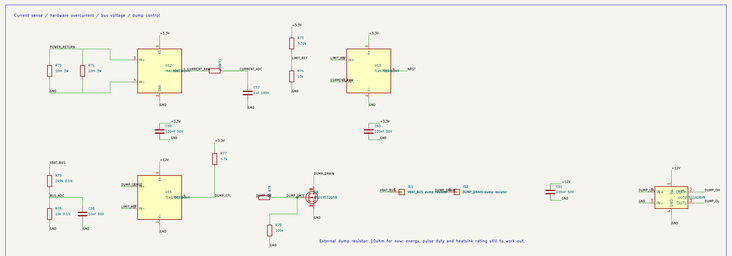
100V MOSFETs: local ON/ switching diodes/ prototype 20kHz PWM

Hardware enable: PAD + reset + emergency loop + phase requests



12V supervisor threshold nominal 11.50V; span CT delay ~20ms

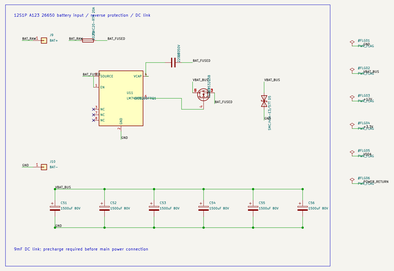
Current sense / hardware overcurrent / bus voltage / dump control



External dump resistor; 10ohm for now; energy, pulse duty and heatsink rating still to work out.

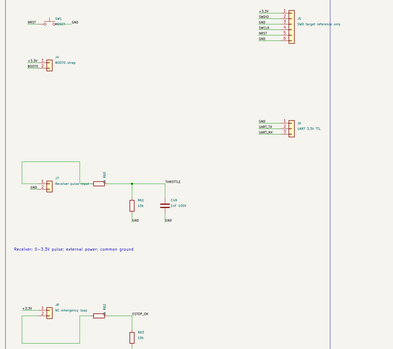
Dump chopper: hysteresis, stability and pulse energy still to check.

125SP A123 26650 battery input / reverse protection / DC link

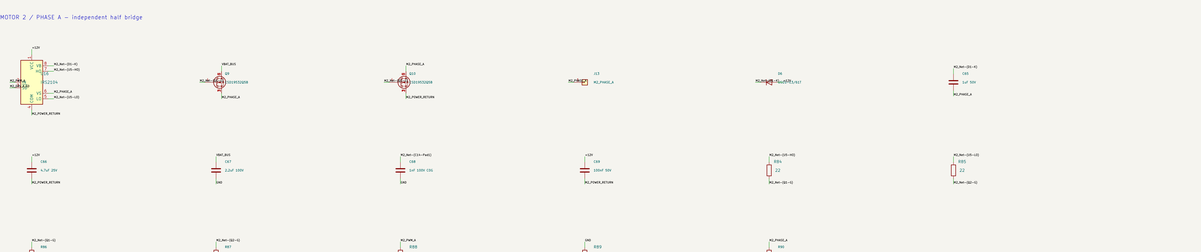


8mF DC link; precharge required before main power connection

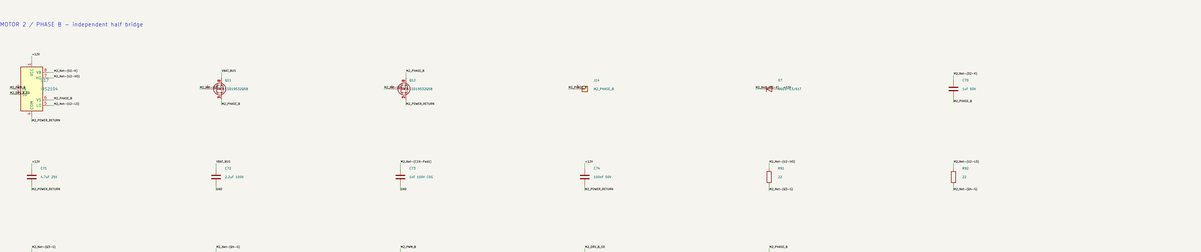
SWD / UART / receiver / emergency-stop interfaces



MOTOR 2 / PHASE A — independent half bridge



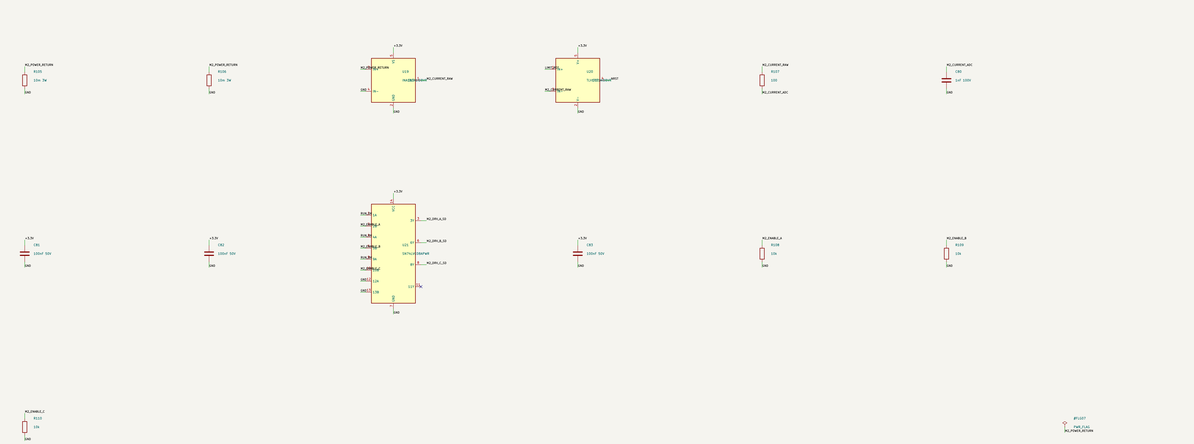
MOTOR 2 / PHASE B — independent half bridge



MOTOR 2 / PHASE C — independent half bridge



MOTOR 2 / Independent single-shunt feedback and common hardware shutdown



TARGET: 125SP A123 26650 + two Boosted Board V1 motors. Exact cell revision and motor ratings still to confirm.

MCU: TIM3 PA6/PA7/PB0 PWM; PB6/PB7/PB8 enables; PB1 ADC9 current; TIM1 remains motor1, No firmware yet.

Each bridge: 2 x 10mOhm parallel, gain20 => 0.100V/A; nominal trip 2A…775A. Overcurrent trip threshold only, not a set motor current limit.

Single low-side shunt; synchronous sampling/reconstruction required; blind switching states and reverse/regen current not covered by INA180.

Shared 20A fuse is not an electronic battery limiter. Pack/BMS charge, discharge, cold-temperature and cell imbalance limits remain required.

8mF at%3.2V stored6.29Wi. External precharge/contact sequencing needed. No automatic precharge on the board.

LMT65504DV recommended /50V absolute max%3.2V steady state compatible; SMCJ58 72.7V clamp is above that limit. Keep regen off until BMS and energy handling are sorted.

Patrick Liu, Marco Strassia
Electric_Bike
File: MotorController-Rebuild.kicad_sch

Title: UltimateVroom dual-motor controller

Size: A3	Date: 2026-05-29	Rev: A
KiCad 9.0.6		Id: 1/1